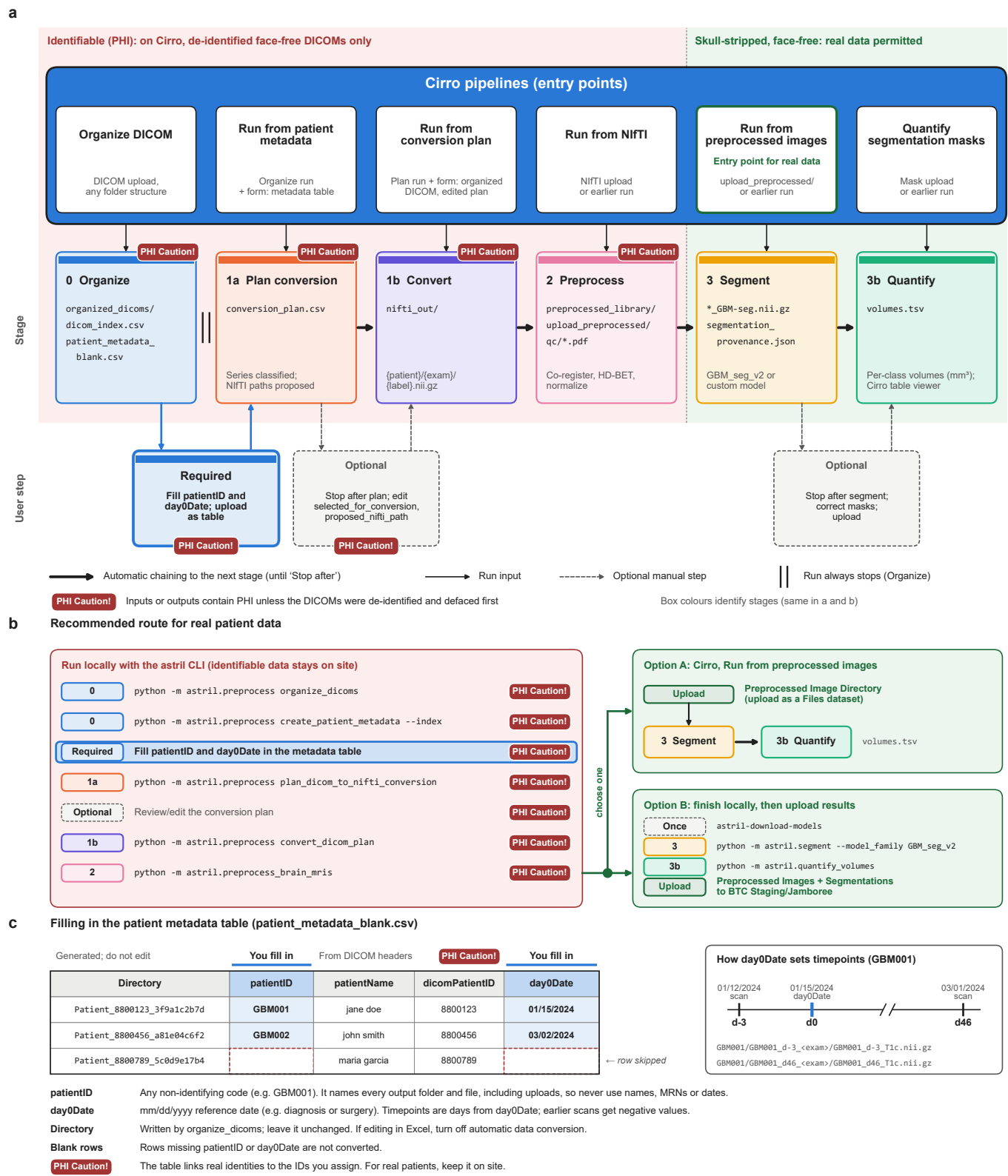

a
Identifiable (PHI): on Cirro, de-identified face-free DICOMs only
Skull-stripped, face-free: real data permitted
Cirro pipelines (entry points)
Organize DICOM
DICOM upload, any folder structure
Run from patient metadata
Organize run + form: metadata table
Run from conversion plan
Plan run + form: organized DICOM, edited plan
Run from NIfTI
NIfTI upload or earlier run
Run from preprocessed images
Entry point for real data
upload_preprocessed/ or earlier run
Quantify segmentation masks
Mask upload or earlier run
Stage
PHI Caution!
0 Organize
organized_dicoms/
dicom_index.csv
patient_metadata_blank.csv
1a Plan conversion
conversion_plan.csv
Series classified; NIfTI paths proposed
1b Convert
nifti_out/
{patient}/{exam}/{label}.nii.gz
2 Preprocess
preprocessed_library/
upload_preprocessed/
qc/*.pdf
Co-register, HD-BET, normalize
3 Segment
*_GBM-seg.nii.gz
segmentation_provenance.json
GBM_seg_v2 or custom model
3b Quantify
volumes.tsv
Per-class volumes (mm³); Cirro table viewer
User step
Required
Fill patientID and day0Date; upload as table
Optional
Stop after plan; edit selected_for_conversion, proposed_nifti_path
Optional
Stop after segment; correct masks; upload
Automatic chaining to the next stage (until 'Stop after')
Run input
Optional manual step
Run always stops (Organize)
PHI Caution! Inputs or outputs contain PHI unless the DICOMs were de-identified and defaced first
Box colours identify stages (same in a and b)
b
Recommended route for real patient data
Run locally with the astril CLI (identifiable data stays on site)
0 python -m astril.preprocess organize_dicoms
0 python -m astril.preprocess create_patient_metadata --index
Required Fill patientID and day0Date in the metadata table
1a python -m astril.preprocess plan_dicom_to_nifti_conversion
Optional Review/edit the conversion plan
1b python -m astril.preprocess convert_dicom_plan
2 python -m astril.preprocess_brain_mris
choose one
Option A: Cirro, Run from preprocessed images
Upload
Preprocessed Image Directory (upload as a Files dataset)
3 Segment
3b Quantify
volumes.tsv
Option B: finish locally, then upload results
Once astril-download-models
3 python -m astril.segment --model_family GBM_seg_v2
3b python -m astril.quantify_volumes
Upload
Preprocessed Images + Segmentations to BTC Staging/Jamboree
c
Filling in the patient metadata table (patient_metadata_blank.csv)
Generated; do not edit
You fill in
From DICOM headers
PHI Caution!
You fill in
Directory
patientID
patientName
dicomPatientID
day0Date
Patient_8800123_3f9a1c2b7d
GBM001
jane doe
8800123
01/15/2024
Patient_8800456_a81e04c6f2
GBM002
john smith
8800456
03/02/2024
Patient_8800789_5c0d9e17b4
maria garcia
8800789
← row skipped
How day0Date sets timepoints (GBM001)
01/12/2024 scan
01/15/2024 day0Date
03/01/2024 scan
d-3
d0
d46
GBM001/GBM001_d-3_<exam>/GBM001_d-3_T1c.nii.gz
GBM001/GBM001_d46_<exam>/GBM001_d46_T1c.nii.gz
patientID Any non-identifying code (e.g. GBM001). It names every output folder and file, including uploads, so never use names, MRNs or dates.
day0Date mm/dd/yyyy reference date (e.g. diagnosis or surgery). Timepoints are days from day0Date; earlier scans get negative values.
Directory Written by organize_dicoms; leave it unchanged. If editing in Excel, turn off automatic data conversion.
Blank rows Rows missing patientID or day0Date are not converted.
PHI Caution! The table links real identities to the IDs you assign. For real patients, keep it on site.